Table 4-1. Design Standards Based on Airplane Design Group (ADG)

Item	ADG					
	I	II	III	IV	V	VI
Taxiway and Taxilane Protection						
TSA (maximum ADG wingspan)	49 ft (14.9 m)	79 ft (24.1 m)	118 ft (36.0 m)	171 ft (52.1 m)	214 ft (65.2 m)	262 ft (79.9 m)
TOFA [2]	89 ft (27.1 m)	124 ft (37.8 m)	171 ft (52.1 m)	243 ft (74.1 m)	285 ft (86.9 m)	335 ft (102.1 m)
TLOFA [2]	79 ft (24.1 m)	110 ft (33.5 m)	158 ft (48.2 m)	224 ft (68.3 m)	270 ft (82.3 m)	322 ft (98.1 m)
Taxiway and Taxilane Separation						
Taxiway centerline to parallel taxiway centerline [1]	70 ft (21.3 m)	101.5 ft (30.9 m)	144.5 ft (44.0 m)	207 ft (63.1 m)	249.5 ft (76.1 m)	298.5 ft (91.0 m)
Taxiway centerline to fixed or movable object [2]	44.5 ft (13.6 m)	62 ft (18.9 m)	85.5 ft (26.1 m)	121.5 ft (37.0 m)	142.5 ft (43.4 m)	167.5 ft (51.1 m)
Taxilane centerline to parallel taxilane centerline [1]	64 ft (19.5 m)	94.5 ft (28.8 m)	138 ft (42.1 m)	197.5 ft (60.2 m)	242 ft (73.8 m)	292 ft (89.0 m)
Taxilane centerline to fixed or movable object [2]	39.5 ft (12.0 m)	55 ft (16.8 m)	79 ft (24.1 m)	112 ft (34.1 m)	135 ft (41.1 m)	161 ft (49.1 m)
Wingtip Clearance						
Taxiway wingtip clearance	20 ft (6.1 m)	22.5 ft (6.9 m)	26.5 ft (8.1 m)	36 ft (11.0 m)	35.5 ft (10.8 m)	36.5 ft (11.1 m)
Taxilane wingtip clearance	15 ft (4.6 m)	15.5 ft (4.7 m)	20 ft (6.1 m)	26.5 ft (8.1 m)	28 ft (8.5 m)	30 ft (9.1 m)

Note 1: See Figure 4-5.
Note 2: See Figure 4-6.
Note 3: See paragraphs 4.5.3.1 and 4.5.4.1 for TSA and TOFA standards at fillets.

Table 4-2. Design Standards Based on Taxiway Design Group (TDG)

Item	TDG							
	1A	1B	2A	2B	3	4	5	6
Taxiway/Taxilane Width [2]	25 ft (7.6 m)	25 ft (7.6 m)	35 ft (10.7 m)	35 ft (10.7 m)	50 ft (15.2 m)	50 ft (15.2 m)	75 ft (22.9 m)	75 ft (22.9 m)
Taxiway Edge Safety Margin [1]	5 ft (1.5 m)	5 ft (1.5 m)	7.5 ft (2.3 m)	7.5 ft (2.3 m)	10 ft (3.0 m)	10 ft (3.0 m)	14 ft (4.3 m)	14 ft (4.3 m)
Taxiway Shoulder Width [3]	10 ft (3.0 m)	10 ft (3.0 m)	15 ft (4.6 m)	15 ft (4.6 m)	20 ft (6.1 m)	20 ft (6.1 m)	30 ft (9.1 m)	30 ft (9.1 m)

Note 1: See Figure 4-4.
Note 2: It is acceptable to reduce taxilane widths for TDG 1A and 1B to 20 feet (6 m).
Note 3: When the most demanding aircraft has four engines and is TDG 6, the standard taxiway shoulder width is 40 feet (12.2 m).

1. For locations where the aircraft is under ATC ground control at push back, a non-movement area boundary marking may serve as the vehicle limit line.
2. For locations where the non-movement area boundary marking is remote from the gate area, consider placement of a vehicle limit line marking per the guidelines of Airports Council International (ACI), https://airportscouncil.org/, publication *Apron Markings and Signs* to visually define the near edge of an apron TLOFA.

5.6.2.2 **Parking Position.**

Provide sufficient clearance to limit risk of conflict between aircraft (wingtips and fuselage) and adjacent objects while entering or exiting the parking position. Refer to Table 5-1 for recommended minimum clearance dimensions. Increase clearance dimensions based on space necessary for personnel and equipment (GSE and ARFF) to operate safely around the perimeter of the parked aircraft.

1. Additional factors may influence increasing the clearance values such as an operator's GSE work area envelope, equipment storage, and the slope of the passenger boarding bridge ramp for conformance with the Americans with Disabilities Act (ADA) (42 U.S.C. § 12101, et al.).
2. An acceptable level of safety may be achievable with reduced clearance values if controls are in place that minimize the risk of conflict between aircraft and objects (e.g., gate operations rules, GSE management plan, training, etc.).
3. For parking positions parallel to terminal building structures, consider a minimum clearance of 45 feet (13.7 m) to accommodate equipment operations between the wingtip and the building.
4. Refer to Figure 5-2 for an illustration of parking position clearance for passenger terminals.
5. Refer to Figure E-3, Figure E-4, and Figure E-5 for illustrations of parking positions for GA aircraft.

Table 5-1. Parking Position Clearance.

Airplane Design Group (ADG)	Recommended Minimum Clearances
I and II	10 ft (3 m)
III, IV, V, and VI	25 ft (7.6 m)

2. Segregate parking areas for small aircraft (e.g., ADG I) from larger aircraft (e.g., ADG II) to optimize utility and efficiency of apron space.
3. Design separate apron areas to accommodate the critical aircraft intended to use the segment of apron.
4. Account for the effects of jet blast and propeller wash on adjacent aircraft and facilities, per guidelines of paragraph 5.17.

E.2.2 Parking Position.

A parking position represents a location where aircraft can park without impeding other aircraft when parking, and aircraft taxi operations on adjacent taxilanes.

E.2.2.1 **Markings.**

Parking positions consist of marked locations (e.g., tiedowns and lead-in lines) or unmarked positions.

1. Marked parking positions provide a measure of safety for aircraft maneuvering but reduces flexibility on the size of aircraft able to park at the position.
2. Unmarked positions:
   a. Provide flexibility to accommodate a diverse mix of aircraft types.
   b. Requires judgmental maneuvering and possibly assistance from a parking marshal or wing-walkers.

E.2.2.2 **Parking Position Sizing.**

To optimize space, establish separate parking areas for the different groupings of aircraft anticipated to use the airport. Table E-1 provides area values based on the common tiedown size groupings of paragraph E.2.3.3, plus clearance values of Table 5-1. Airport operators may apply these values or establish custom values based on specific aircraft dimensions using its airport.

Table E-1. Minimum Parking Position Sizing

Wingspan	Length	Area
49 ft (14.9 m)	< 30 ft (9.1 m)	2,065 sf (192 m^2)
49 ft (14.9 m)	30 to 45 ft (9.1 m to 13.7 m)	2,950 sf (274 m^2)
79 ft (24.1 m)	45 to 60 ft (13.7 m to 18.3 m)	5,785 sf (537 m^2)
79 ft (24.1 m)	60 to 75 ft (18.3 m to 22.9 m)	7,120 sf (661 m^2)
79 ft (24.1 m)	75 to 90 ft (22.9 m to 27.4 m)	8,455 sf (786 m^2)

Note 1: Values for area are calculated by multiplying the sum of the wingspan dimension plus 10 feet (3 m) by the sum of the maximum aircraft length plus 5 feet (1.5 m) and rounding up to the nearest multiple of 5.

Note 2: Values for area do not include the TLOFA in front of the parking position.

E.2.3 Tiedowns.

A tiedown is a distinct type of parking stand designed to accommodate a specific range of aircraft sizes.

E.2.3.1 **Benefits.**

Securing aircraft reduces risk of damage to parked aircraft due to wind forces from weather events (e.g., storms, wind gusts, microbursts, etc.) and mechanical forces (e.g., jet blast, propeller wash). Unsecured parking locations expose aircraft to risk of damage from wind forces by becoming airborne, spun around, or flipped over.

E.2.3.2 **Tiedown Characteristics.**

1. The three-point method is a common tiedown involving two tiedown anchors along the wing axis and one anchor at the tail of the aircraft.
2. Commonly includes a 6-inch (152 mm) yellow striped marking in the shape of a tee to demonstrate the location for wing axis and tail location.
3. Tiedown locations typically occur on paved aprons but may also occur in turf areas (e.g., overflow parking or tail tiedown anchor located off edge of paved apron).

E.2.3.3 **Tiedown Area Guidelines.**

The sizing of tiedown parking locations at an airport is a function of the aircraft using the airport. Base tiedown area dimensions on aircraft wingspan and length. To determine suitable tiedown area dimensions, each airport operator needs to assess aircraft activity on its apron. The following guidelines show common aircraft groupings based on aircraft lengths for an airport operator to consider for its airport.

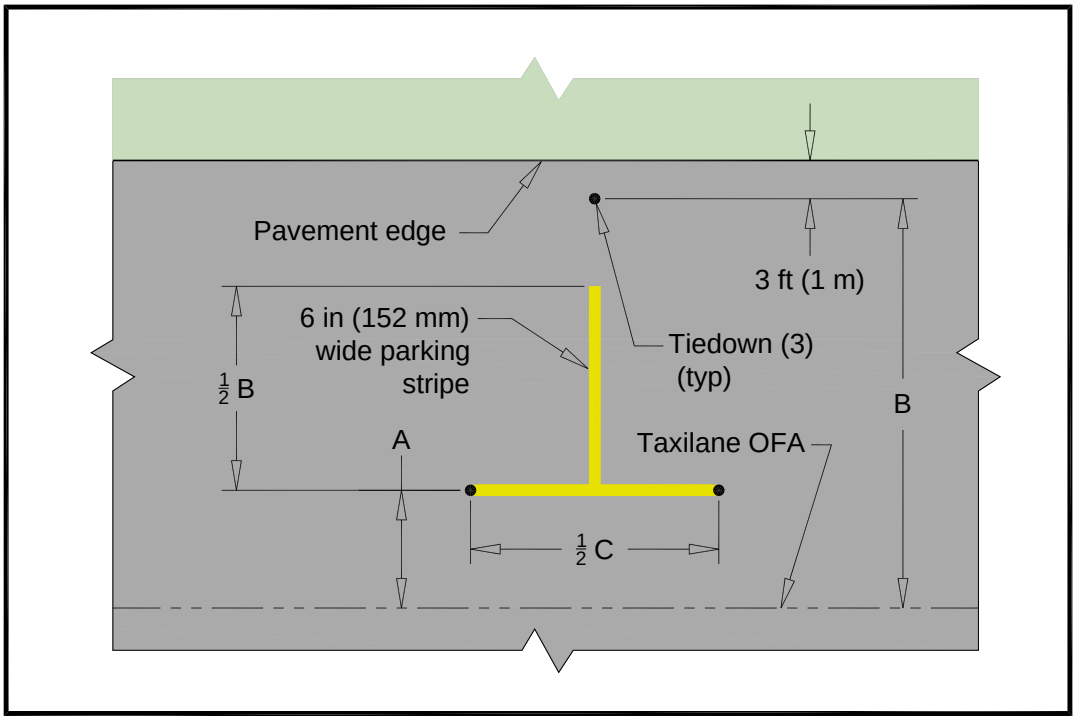
1. ADG I aircraft
   a. 30 ft (9.1 m) aircraft length
   b. 45 ft (13.7 m) aircraft length
2. ADG II aircraft
   a. 60 ft (18.3 m) aircraft length
   b. 75 ft (22.9 m) aircraft length
   c. 90 ft (27.4 m) aircraft length
3. ADG III aircraft
   a. 100 ft (30.5 m) aircraft length
   b. 115 ft (35 m) aircraft length

E.2.3.4 **Tiedown Anchor Layout.**

Refer to Figure E-1 for an example of a tiedown layout. Airport operators may customize the tiedown layout dimensions to address local needs. Consider providing separate groupings of different tiedown layouts to accommodate the varying aircraft sizes that use the airport.

1. Provide two wing tiedown anchors at the end of the marked stripe spaced at least half the wingspan of the critical aircraft.
2. From the edge of the TLOFA:
   a. Locate wing tiedown anchors at a distance so the nose of aircraft using the tiedown does not penetrate the adjacent TLOFA.
   b. Locate the tail tiedown anchor from the TLOFA at a distance not less than the length of the critical aircraft.
3. Ensure no part of the tiedown anchor protrudes above the pavement surface; or no more than 1 inch (25 mm) above surrounding turf.

Figure E-1. Tiedown Layout



Note 1: Figure keys:

A. Distance from main gear to nose of critical aircraft assuming nose of aircraft at the outer edge of the TLOFA.
B. Overall length of the critical aircraft.
C. Wingspan of the critical aircraft.

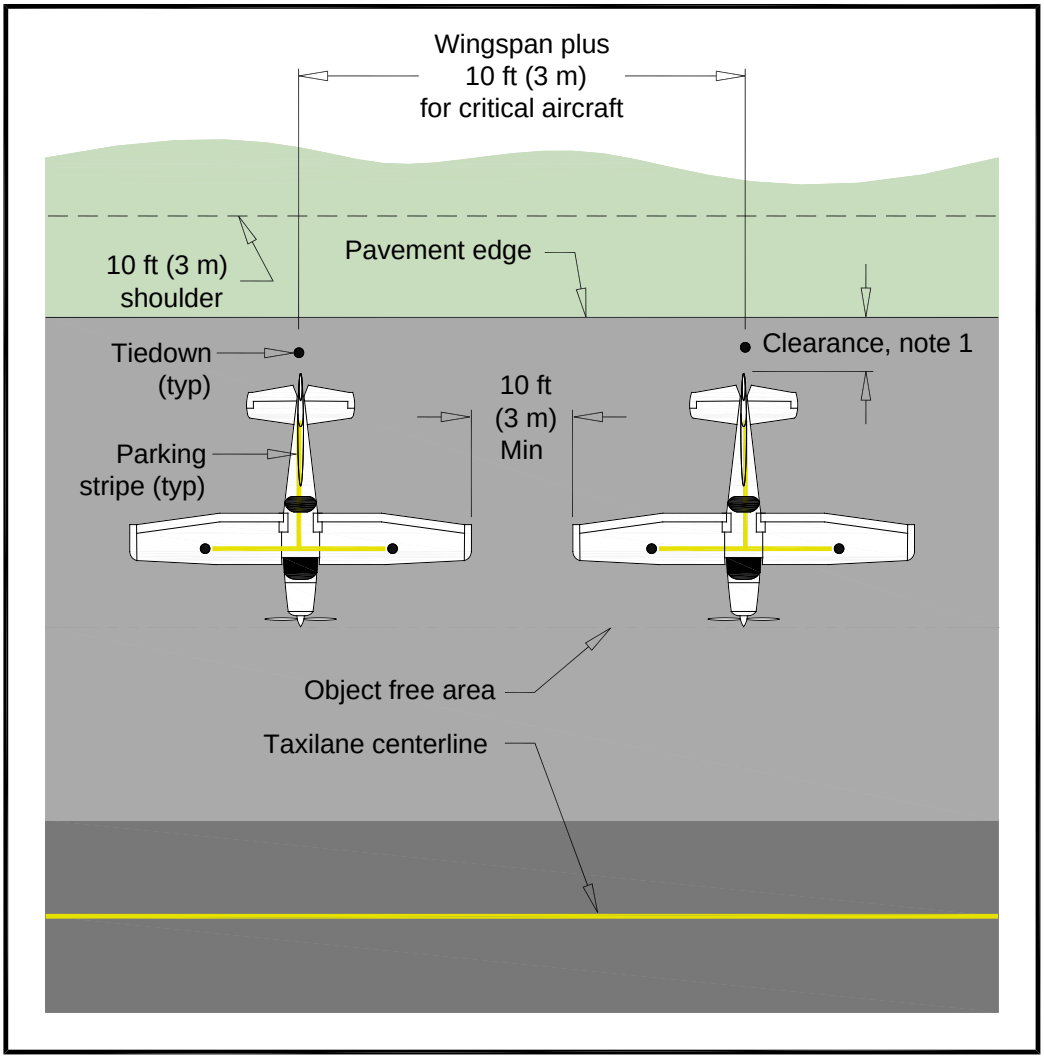
Note 2: Refer to paragraph E.2.3 for tiedown design considerations.
Note 3: Ensure the aircraft using the tiedown does not violate the TLOFA.
Note 4: Take into account the length of tiedown ropes and the risk to propeller entanglement.

E.2.4 Apron Layout Considerations.

E.2.4.1 **General.**

1. Refer to Figure E-2 for an illustration of tiedown spacing relationship.

Figure E-2. Tiedown Spacing

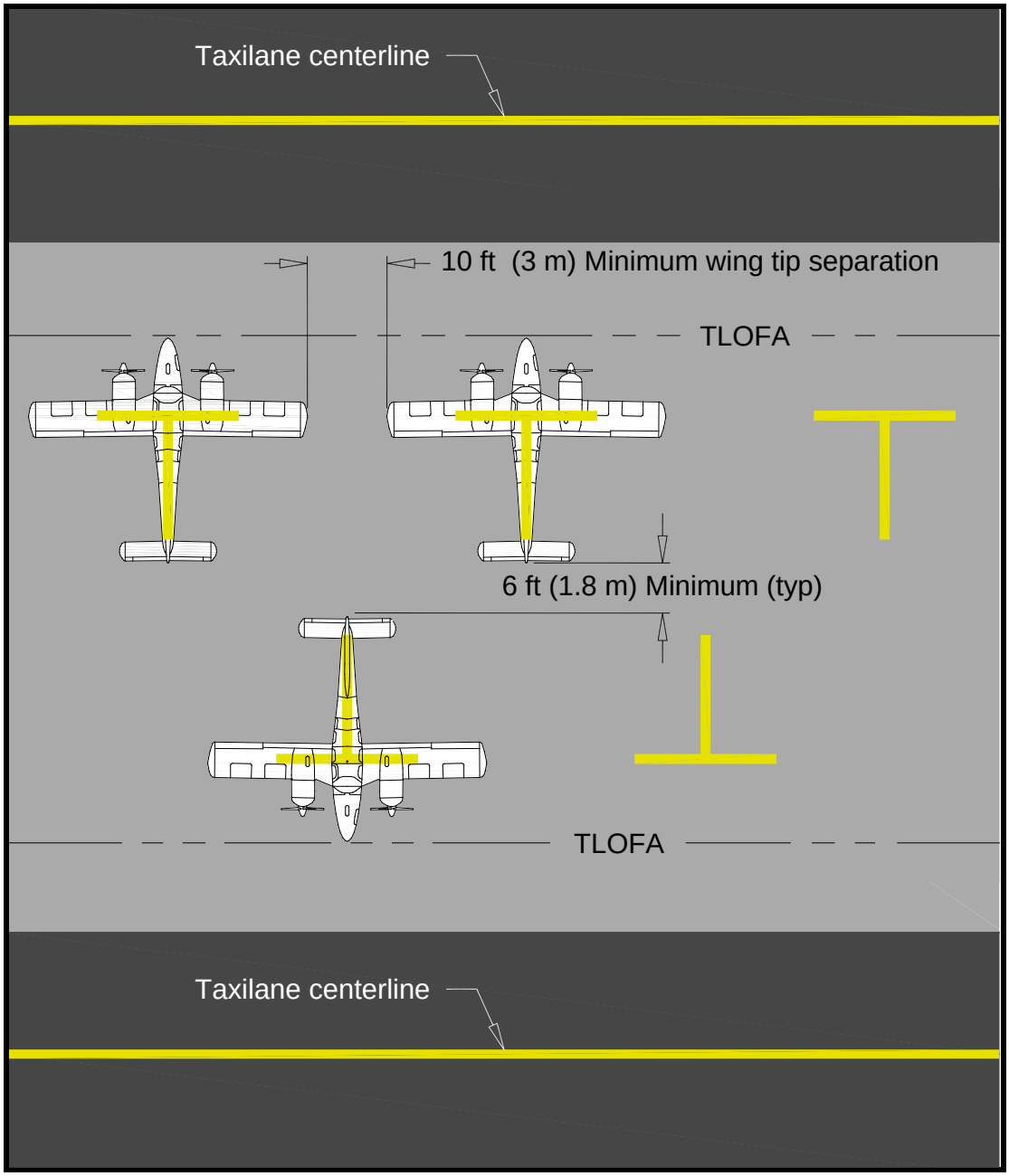


Note 1: Provide sufficient space between the pavement edge and the tail to permit pilots to walk around aircraft during preflight check.
Note 2: Layout tiedowns so that no part of a parked aircraft violates a TLOFA, TOFA, or ROFA.
Note 3: Depth of tiedown position is dependent on critical aircraft. Refer to paragraph E.2.3.3.
Note 4: Ensure no part of parked aircraft violates a TOFA or TLOFA.
Note 5: Maintain recommended obstruction clearances, per Table 5-1.
Note 6: Orient parking positions to align with the direction of prevailing winds at the airport.
Note 7: Group parking positions for similar aircraft wingspan and length to facilitate efficient layout of the apron.
Note 8: Incorporate flexibility for infrequent operations by large aircraft (e.g., ADG-III) by designating and marking an area capable of accommodating both large and medium (ADG-II) size aircraft.

E.2.4.2 **Nested Parking Positions.**

Applying a nested parking position configuration optimizes available apron space and shortens the length of the associated taxilanes. Refer to Figure E-3 for a typical nested parking position layout. Nested parking positions also offer flexibility by accommodating both small and large aircraft.

Figure E-3. Nested Tiedowns



Note 1: Maintain clearance of TLOFA.
Note 2: Tighter clearances may be suitable when aircraft enter and exit a parking position by tug.